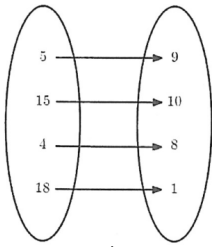
# Functions and Transformations

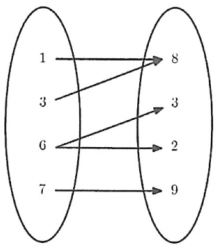
Questions 1–175

**2.10 — Question 1.** Identify which of the following mapping diagrams depict a function:

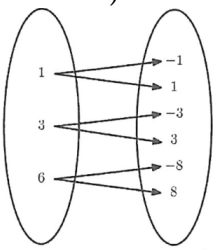
**a)**



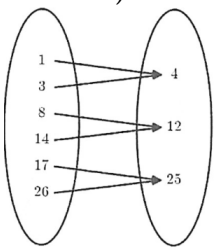

**b)**




**c)**

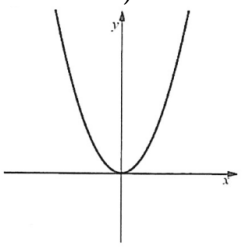



**d)**

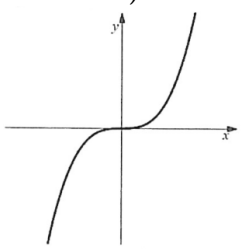



**2.10 — Question 2.** For each of the following, state whether it is one-to-one, many-to-one, one-to-many or many-to-many and if it is function or not:

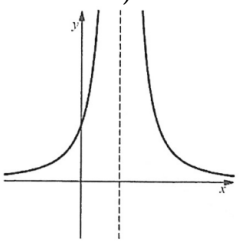
**a)**



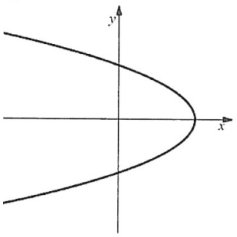

**b)**



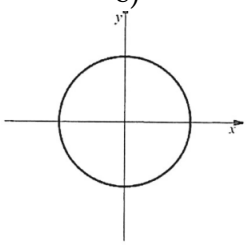

**c)**



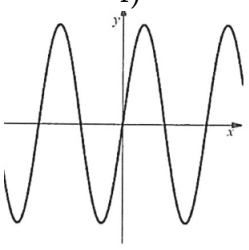

**d)**




**e)**




**f)**



**2.10 — Question 3.** It is given that $f : x \mapsto 5x - 3$ where $x \in \{1, 2, 3, 4\}$.

a) Represent $f$ on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

**2.10 — Question 4.** It is given that $f(x) = 3x^2 - 1$ where $x \in \{-2, -1, 0, 1\}$.

a) Represent $f(x)$ on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

**2.10 — Question 5.** It is given that $f : x \mapsto \sqrt{x-1}$ where $x \in \{3, 5, 8, 10\}$.

a) Represent $f(x)$ on a mapping diagram.

b) State whether $f(x)$ is a function and if so, what type of function it is.

**2.10 — Question 6.** Determine which of these functions are one-to-one and which are many to one:

a) $f(x) = 4x - 3,\ \{x : x \in \mathbb{R}\}$

b) $f : x \mapsto x^2 + 5,\ \{x : x \in \mathbb{R}\}$

c) $f(x) = \frac{1}{x},\ \{x : x \in \mathbb{R}, x \neq 0\}$

d) $f : x \mapsto 2x^3,\ \{x : x \in \mathbb{R}\}$

e) $f(x) = x(x + 4),\ \{x : x \in \mathbb{R}\}$

f) $f : x \mapsto 3\cos x,\ \{x : x \in \mathbb{R}, 0° \leq x \leq 180°\}$

g) $f(x) = (x - 7)^2 + 3,\ \{x : x \in \mathbb{R}, x > 7\}$

h) $f : x \mapsto x^2(x - 4),\ \{x : x \in \mathbb{R}, x > 0\}$

i) $f(x) = 3\ln x,\ \{x : x \in \mathbb{R}, x > 0\}$

j) $f : x \mapsto \sqrt{x},\ \{x : x \in \mathbb{R}, x \geq 4\}$

**2.10 — Question 7. The answers for the following questions which require a range of values should be given in set notation** State the range of the following, given that in each case $x \in \mathbb{R}$:

a) $f(x) = 1 + 2^{-x}$

b) $y = (x + 3)^2 - 5$

c) $f(x) = x(x - 2)^2$

d) $y = -2e^{-4x}$

e) $f(x) = 5 - 3^{-x}$

**2.10 — Question 8.** For each of the following functions $y = f(x)$ below, state the largest possible domain and the corresponding range of $f$:

a) $y = \cos x$

b) $y = \sqrt{x}$

c) $y = -(x + 4)^2 - 9$

d) $y = 2x^2 - 6x + 9$

e) $y = \frac{1}{x} + 4$

f) $y = \frac{1}{x-3}$

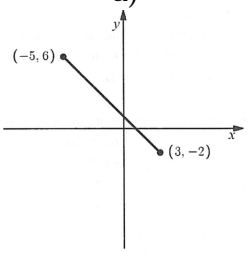
g) $y = 2 + e^x$

h) $y = 4 \ln x$

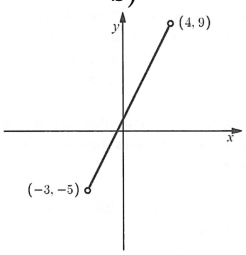
i) $y = 3 \ln(x - 2)$

**2.10 — Question 9.** For each of the graphs below, state the domain and the range:

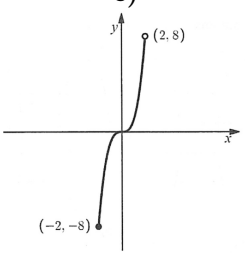
**a)**



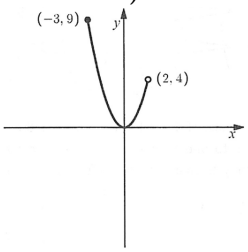

**b)**



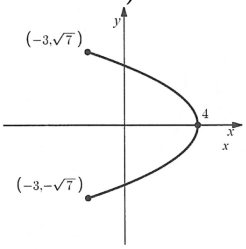

**c)**



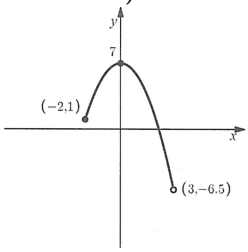

**d)**




**e)**

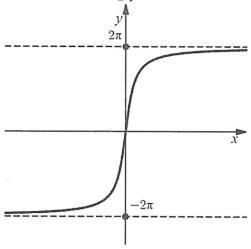



**f)**

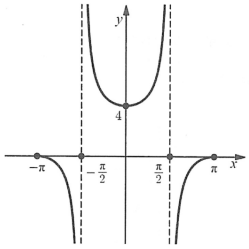



**2.10 — Question 9 (continued).** For each of the graphs below, state the domain and the range:

**g)**



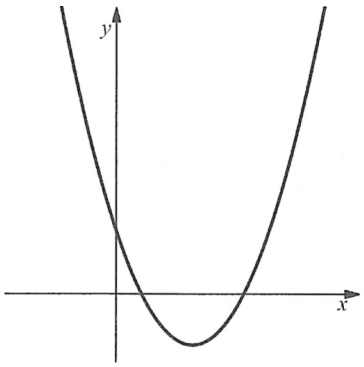

**h)**



**2.10 — Question 10.** Sketch a graph which is many-to-one, has domain $\{x : x \in \mathbb{R}\}$, range $\{y : y \in \mathbb{R}, y \leq 3\}$, the $y$-intercept is negative, and its only maximum occurs at $x = p$ where $p > 0$.

**2.10 — Question 11.** Sketch a graph which is one-to-one, has domain $\{x : x \in \mathbb{R}\}$, range $\{y : y \in \mathbb{R}\}$, the $y$-intercept is positive and it has a single stationary point which occurs at $x = p$ where $p > 0$.

**2.10 — Question 12.** The graph below shows $f(x) = \frac{1}{2}(x-3)^2 - 2$.



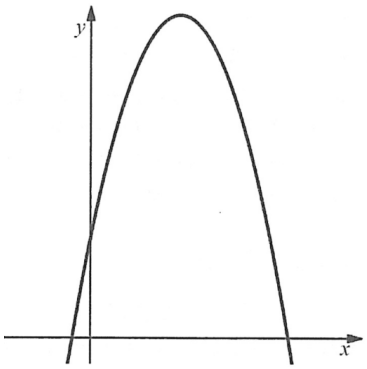

Find the range of $f(x)$ when the domain is restricted as follows:

a) $x \geq 5$

b) $x \leq 2$

c) $0 \leq x \leq 4$

**2.10 — Question 13.** The graph below shows $f(x) = -0.5x^2 + 5x + 5.5$.




Find the range of $f(x)$ when the domain is restricted as follows:

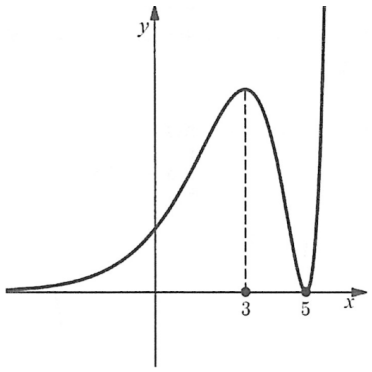
a) $x \geq 5$

b) $x \leq 2$

c) $0 < x < 4$

d) $3 \leq x \leq 7$

**2.10 — Question 14.** The graph below shows $f(x) = \frac{1}{12}(x-5)^2 e^x$.




Find the range of $f(x)$ when the domain is restricted as follows:

a) $x < 5$

b) $3 \leq x \leq 5$

c) $x \geq 5$

d) $0 \leq x < 3$

**2.10 — Question 15.** Given $f : x \mapsto x^2 + 8x + 4$.

a) Sketch the graph of $y = f(x)$, labelling any intersections with the axes and any stationary points.

b) State its range for $x \in \mathbb{R}$.

**2.10 — Question 16.** Find the range of each of the following functions:

a) $f(x) = 4x + 1,\ \{x : x \in \mathbb{R}, x \geq 1\}$

b) $f : x \mapsto \frac{1}{3}x - 2,\ \{x : x \in \mathbb{R}, x < -2\}$

c) $f(x) = 5 - 8x,\ \{x : x \in \mathbb{R}, -2 \leq x < 4\}$

d) $f(x) = 2\sqrt{x},\ \{x : x \in \mathbb{R}, 4 < x \leq 16\}$

e) $f : x \mapsto 3x - x^2,\ \{x : x \in \mathbb{R}, 0 < x < 3\}$

f) $f(x) = \frac{1}{1+x^2},\ \{x : x \in \mathbb{R}\}$

**2.10 — Question 17.** Given that for $x \in \mathbb{R}$, $f(x) = \dfrac{1}{2x^2 - 4x + 9}$. By first completing the square on $2x^2 - 4x + 9$ state the range of $f$.

**2.10 — Question 18.** The function $f(x) = 2x^2 + kx + 13$, where $k$ is a constant, has domain $x \in \mathbb{R}$. It is known that the range of $f$ is $f(x) \geq -5$. Find the possible value(s) of $k$.

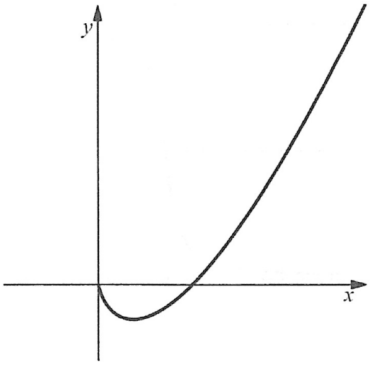
**2.10 — Question 19.** The function $f(x) = k(x-a)(x-b)$ where $a$, $b$, $k$ are all positive, is restricted in its domain such that it is one-to-one and $x \geq p$ where $p$ gives the largest possible range. Find the value of $p$ giving your answer in terms of $a$ and $b$.

**2.10 — Question 20.** Find the largest possible domain and range of

a) $f : x \mapsto -x^4 + 4x$

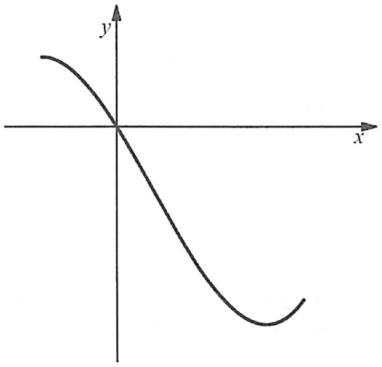
b) $f : x \mapsto 4x - e^{2x}$

**2.10 — Question 21. Prerequisite knowledge for Q21: Differentiation of standard functions and product rule** The graph below shows $f(x) = x \ln x$.




Find the largest possible domain and range of $f : x \mapsto x \ln x$.

**2.10 — Question 22.** The graph below shows $f(x) = x^3 - 3x^2 - 24x$ for $-2 < x < 5$.




Find the range of

a) $f$

b) $g$ given that $g(x) = 2f(x)$

c) $h$ given that $h(x) = 3f(x) - 4$

**2.10 — Question 23.** Given $f : x \mapsto 5x - 1$ and $g : x \mapsto x^2 + 2$, find

a) $fg(-1)$

b) $gf(2)$

c) $ff(4)$

**2.10 — Question 24.** Given $f(x) = x^2$ and $g(x) = \dfrac{1}{x-2}$, find $fg(4)$.

**2.10 — Question 25.** Given $f(x) = 2x + 1$ and $g(x) = x^2$, find

a) $fg(x)$

b) $gf(x)$

c) $ff(x)$

d) the value of $a$ such that $fg(a) = 33$ and $a$ is positive

e) the values of $x$ such that $fg(x) = gf(x)$

**2.10 — Question 26.** Given $f(x) = \frac{1}{x}$ find

a) $f^{89}(x)$

b) $f^{100}(x)$

**2.10 — Question 27.** Given $f(x) = 1 + \sqrt{x}$ and $g(x) = 3 - 2^{-x}$. Solve $gf(x) = \frac{23}{8}$.

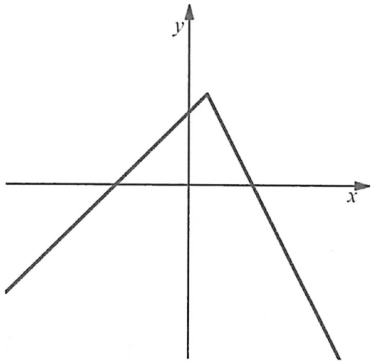
**2.10 — Question 28.** Given $f : x \mapsto 5\log_{10} x$ and $g : x \mapsto 10^x$

a) find $fg(x)$

b) show that $fgf(x) = 25\log_{10} x$

**2.10 — Question 29.** A function is given by $f(x) = -kx^2 + 12x - 11$ where $k$ is a positive constant. It is known that $ff(2) = -2$. Find the possible values of $k$.

**2.10 — Question 30.** The graph below shows $f(x) = \begin{cases} x+4, & x \leq 1 \\ 7-2x, & x > 1. \end{cases}$

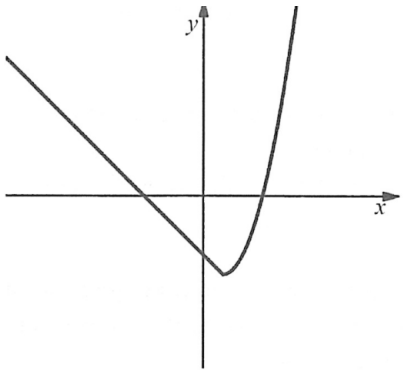



a) State the value of $ff(1)$

b) Find the value of $ff(\frac{1}{2})$

c) Find the range of values of $x$ for which $f(x) < -2$

**2.10 — Question 31.** The graph below shows $f(x) = \begin{cases} -x-3, & x < 1 \\ x^2-2x-3, & x \geq 1. \end{cases}$



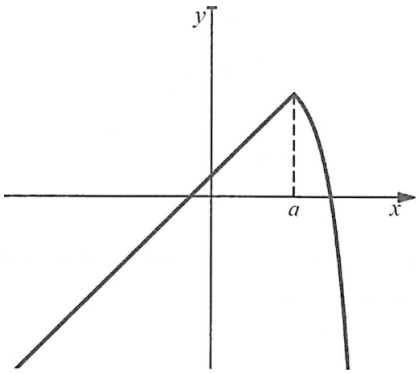

a) State the value of $ff(1)$

b) State the range of $f$

c) Find the value of $ff(2)$

d) Find the range of values of $x$ for which $f(x) < 0$

**2.10 — Question 32.** The graph below shows $f(x) = \begin{cases} x+1, & x < a \\ 6 - e^{x-4}, & x \geq a. \end{cases}$




The range of $f$ is $f(x) \leq 5$. Find

a) the value of $a$

b) the exact value of $ff(\frac{7}{2})$

c) the range of values of $x$ for which $f(x) \geq 0$

**2.10 — Question 33.** Given $f : x \mapsto \dfrac{2x+1}{x-5}$, $x \neq 5$

a) Find the range of $f$

b) Find $ff(x)$

**2.10 — Question 34.** Given $f(x) = \dfrac{ax+1}{x-4}$, $x \neq 4$ where $a$ is a constant and $a \neq -\frac{1}{4}$

a) Find the range of $f$

b) Find $ff(x)$

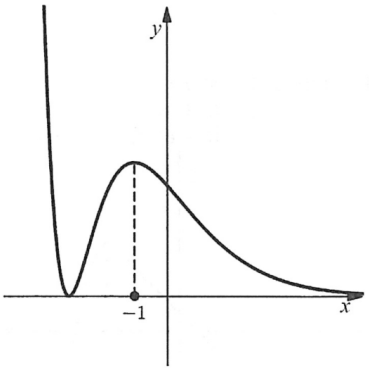
**2.10 — Question 35.** Given $f(x) = x^2 - ax$ and $g(x) = a^2 - x$ where $a$ is a positive constant and $fg(2) = 0$

a) find $fg(x)$

b) find the possible values of $a$ giving $a$ in exact form.

**2.10 — Question 36. The answers for the following questions which require a range of values should be given in interval notation**

The graph below shows $f(x) = 2(x+3)^2 e^{-x}$.




Find the range of $f(x)$ when the domain is restricted as follows:

a) $x < -3$

b) $-3 \leq x \leq -1$

c) $x \geq 0$

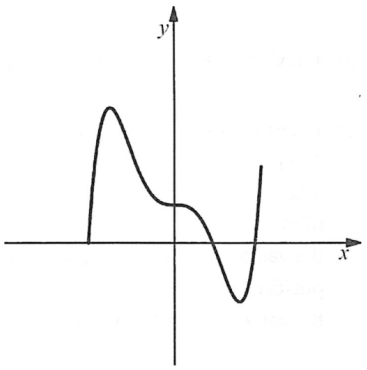
d) $-3 \leq x < 0$

**2.10 — Question 37.** Given $f(x) = 2x^2 - 12x + 13$

a) Sketch the graph of $y = f(x)$, labelling any intersections with the axes and any stationary points.

b) State its range for $x \in \mathbb{R}$.

**2.10 — Question 38.** The graph below shows $f(x) = 16x^5 - 15x^3 + 1$ for $-1 < x < 1$.




Find the range of

a) $f$

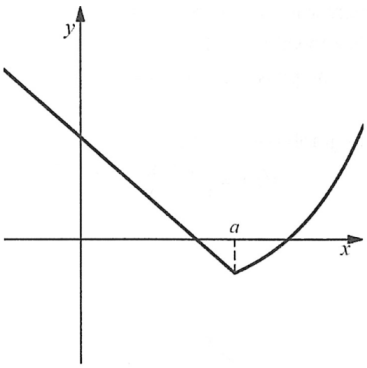
b) $g$ given that $g(x) = 64f(x)$

c) $h$ given that $h(x) = 8f(x) - 10$

Find the domain of

d) $k$ given that $k(x) = f(2x)$

**2.10 — Question 39.** The graph below shows $f(x) = \begin{cases} 3 - kx, & x < a \\ e^{x-2} - 2, & x \geq a \end{cases}$ where $k$ is a positive constant.




The range of $f$ is $f(x) \geq -1$.

a) Find the value of $a$

b) Find the value of $k$

c) Find the exact value of $ff(3)$

d) Find the range of values of $x$ for which $f(x) \leq 0$

**2.11 — Question 1.** Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = 3f(4x)$

b) $y = \frac{1}{2}f(\frac{1}{3}x)$

**2.11 — Question 2.** Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = -f(x + 6)$

b) $y = f(-x) + 6$

**2.11 — Question 3.** Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = \frac{1}{6}f(x-4)$

b) $y = f(8x) - 5$

**2.11 — Question 4. Prerequisite knowledge for Q4: Radians** Describe a sequence of two transformations that will map $y = \sin x$ onto the image:

a) $y = 2\sin(x - \frac{\pi}{3})$

b) $y = \sin(\frac{1}{5}x) + 2$

**2.11 — Question 5.** Describe a sequence of two transformations that will map $y = e^x$ onto the image:

a) $y = 5e^{x/3}$

b) $y = e^{-x} - 9$

**2.11 — Question 6.** Describe a sequence of two transformations that will map $y = \frac{1}{x}$ onto the image:

a) $y = \frac{2}{x-4}$

b) $y = 5 - \frac{1}{x}$

**2.11 — Question 7.**

a) Describe a sequence of two transformations that will map $y = \ln x$ onto the image $y = \ln(2x) - 3$.

b) Describe a **single** transformation that will map $y = \ln x$ onto the image $y = \ln(2x) - 3$.

**2.11 — Question 8.** The point $P(-6, 12)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of $P$ after $y = f(x)$ is mapped to each of these:

a) $y = 2f(3x)$

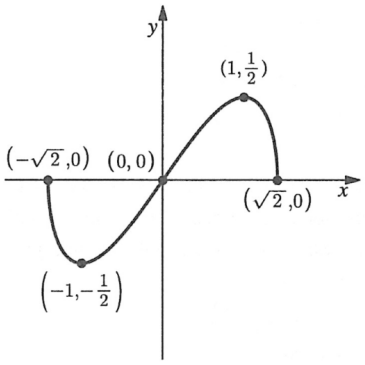
b) $y = -f(x - 5)$

**2.11 — Question 9.** The point $Q(4, -3)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of $Q$ after $y = f(x)$ is mapped to each of these:

a) $y = \frac{1}{3}f(\frac{1}{6}x)$

b) $y = f(-x) + 9$

**2.11 — Question 10.** The graph of $y = f(x)$ is shown in the diagram below.




Sketch the graphs of

a) $y = 2f(\frac{1}{2}x)$

b) $y = -f(x + \sqrt{2})$

c) $y = f(2x) + \frac{1}{2}$

d) $y = f(-x) - 1$

**2.11 — Question 11.** Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = 2f(x) + 3$

b) $y = 9f(x) - 8$

c) $y = \frac{1}{5}f(x) - 10$

d) $y = \frac{3}{2}f(x) + 20$

**2.11 — Question 12.** Describe a sequence of two transformations that will map:

a) $y = 4f(x) - 1$ onto $y = 5f(x) + 2$

b) $y = -f(x) - 5$ onto $y = -\frac{1}{2}f(x) + 7$

**2.11 — Question 13.** Describe a sequence of two transformations that will map $y = f(x)$ onto the image:

a) $y = f(3x + 1)$

b) $y = f(2x - 9)$

c) $y = f(2 - x)$

d) $y = f(-x - 3)$

**2.11 — Question 14.** Describe a sequence of two transformations that will map:

a) $y = f(3x + 8)$ onto $y = f(6x - 4)$

b) $y = f(2 - x)$ onto $y = f(3 - \frac{1}{5}x)$

**2.11 — Question 15.** The point $P(10, -3)$ lies on the graph with equation $y = f(x)$. State the coordinates of the image of $P$ after $y = f(x)$ is mapped to each of these:

a) $y = 2f(x - 5) + 3$

b) $y = -f(x) - 9$

c) $y = f(3x + 5)$

d) $y = f(\frac{1}{4}x - 2)$

**2.11 — Question 16. Prerequisite knowledge for Q16: Radians** Describe a sequence of two transformations that will map $y = \sin x$ onto the image:

a) $y = 4\sin x - 6$

b) $y = \sin(\frac{x}{6} - \frac{\pi}{9})$

c) $y = \cos 2x$

d) $y = 3\cos x$

**2.11 — Question 17.** Describe a sequence of two transformations that will map $y = e^x$ onto the image:

a) $y = e^{6x-10}$

b) $y = e^{9-x}$

c) $y = 10e^x + 20$

d) $y = 18 - e^x$

**2.11 — Question 18.** Describe a sequence of two transformations that will map $y = \frac{1}{x}$ onto the image:

a) $y = \frac{1}{5x-2}$

b) $y = \frac{8}{x} - 9$

c) $y = \frac{6-5x}{x}$

d) $y = \frac{2x+7}{20x}$

**2.11 — Question 19.** Describe a sequence of two transformations that will map $y = \frac{1}{x^2}$ onto the image:

a) $y = -\frac{1}{5x^2}$

b) $y = 1 + \frac{2}{x^2}$

c) $y = \frac{2+3x^2}{6x^2}$

d) $y = \frac{1-6x^2}{8x^2}$

**2.11 — Question 20.**

a) Describe a sequence of two transformations that will map $y = \ln(x + 1)$ onto the image $y = \ln(3x + 3)$.

b) Describe a **single** transformation that will map $y = \ln(x + 1)$ onto the image $y = \ln(3x + 3)$.

**2.11 — Question 21.** Describe a sequence of two transformations that will map $y = 4x + 3$ onto the image:

a) $y = 8x + 7$

b) $y = 2x + 6$

**2.11 — Question 22.** Describe a sequence of two transformations that will map $y = 4 - 5x$ onto the image:

a) $y = -10x - 8$

b) $y = 50x - 40$

**2.11 — Question 23.** Write down the image of $y = 5x + 2$ after a translation by the vector $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$, followed by a stretch, parallel to the $x$-axis, scale factor 2.

**2.11 — Question 24.** Write down the image of $y = 8 - 6x$ after a stretch, parallel to the $y$-axis, scale factor $\frac{1}{3}$, followed by a translation by the vector $\begin{pmatrix} -\frac{2}{3} \\ \frac{4}{3} \end{pmatrix}$.

**2.11 — Question 25.** Describe a sequence of two transformations that will map $y = x^2$ onto the image:

a) $y = 4x^2 + 9$

b) $y = 5 - x^2$

c) $y = \frac{1}{4}(x - 6)^2 + 3$

d) $y = 8(x + 1)^2 - 2$

e) $y = (4x + 1)^2$

f) $y = (\frac{1}{3}x - 9)^2$

g) $y = 2x^2 - 4x - 7$

h) $y = 3x^2 + 12x + 1$

i) $y = -(x + 1)(x - 9)$

j) $y = (6x - 4)(3x + 10)$

**2.11 — Question 26.** Describe a sequence of two transformations that will map $y = x^2 + 2$ onto the image:

a) $y = 9x^2 + 10$

b) $y = 8x^2$

c) $y = 7 - x^2$

**2.11 — Question 27.** Write down the image of $y = x^3$ after a stretch, parallel to the $x$-axis, scale factor $\frac{1}{2}$, followed by a translation by the vector $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

**2.11 — Question 28.** Write down the image of $y = \ln x$ after a stretch, parallel to the $x$-axis, scale factor 9, followed by a translation by the vector $\begin{pmatrix} 6 \\ -\ln 3 \end{pmatrix}$, showing that it can be written in the form $y = \ln(ax + b)$, where $a$ and $b$ are rational numbers.

**2.11 — Question 29.** Find the equation of the curve $y = x^2$ after a reflection in the line:

a) $y = 2$

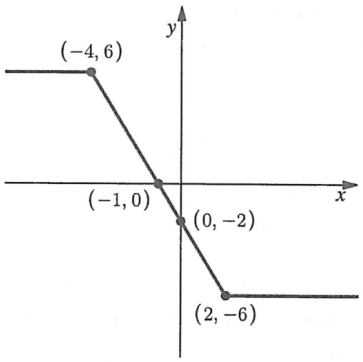
b) $x = 3$

**2.11 — Question 30. Prerequisite knowledge for Q30: Radians** Find the equation of the curve $y = \cos x$ after a reflection in the line:

a) $y = -3$

b) $x = -\frac{\pi}{3}$

**2.11 — Question 31.** The graph of $y = f(x)$ is shown in the diagram below.




Sketch the graphs of

a) $y = f(\frac{1}{2}x - 2)$

b) $y = f(2x + 5)$

c) $y = 3f(x) + 1$

d) $y = \frac{1}{2}f(x) - 4$

**2.11 — Question 32.** Describe the transformation that maps $y = f(x)$ onto the image $x = f(y)$.

**2.11 — Question 33.** Describe the transformation that maps $y = 8 - (x - 3)^2$ onto the image $x = 8 - (y - 3)^2$.

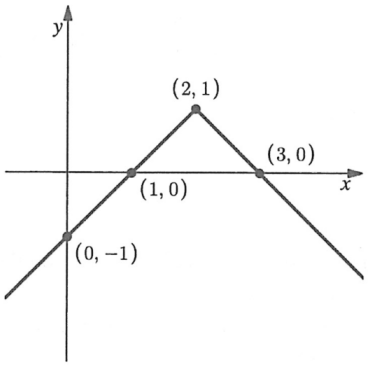
**2.11 — Question 34.** Sketch the image of $(x - 1)^2 + y^2 = 9$ once it has been reflected in the line $y = x$.

**2.11 — Question 35.** Describe a sequence of two transformations that will map $y = \ln x$ onto the image $y = e^{3x}$.

**2.11 — Question 36.** Describe a sequence of two transformations that will map $y = \ln(x - 5)$ onto the image $y = e^x$.

**2.11 — Question 37.** Describe a sequence of extbfthree transformations that will map $y = \ln x$ onto the image $y = 2e^x - 7$.

**2.11 — Question 38.** The graph of $y = f(x)$ is shown in the diagram below.




Sketch the graph of $x = f(y)$.

**2.11 — Question 39.** Describe a sequence of extbffour transformations that will map $y = f(x)$ onto the image $y = 2f(\frac{1}{3}x - 9) - 3$.

**2.11 — Question 40.** Describe a sequence of extbffour transformations that will map $y = f(x)$ onto the image $y = \frac{3}{5}f(\frac{2}{5}x + 4) + 2$.

**2.11 — Question 41.** Find the equation of the curve $y = x^2$ after a reflection in the line $y = -x$.

**2.12 — Question 1.** For each of the following, state whether or not $f^{-1}$ exists:

a) $f(x) = x^3 + 5$ for $x \in \mathbb{R}$

b) $f : x \mapsto \cos x$ for $x \in \mathbb{R}$

c) $f(x) = \tan x$ for $0 \leq x < \pi,\ x \neq \frac{\pi}{2}$

d) $f(x) = x^2 + 5$ for $x \in \mathbb{R}$

e) $f : x \mapsto x^2 + 4x$ for $x \geq -2$

**2.12 — Question 2.** Find the inverse of the following functions:

a) $f(x) = 1 - 2x, \quad x \in \mathbb{R}$

b) $f(x) = \frac{x+3}{4}, \quad x \in \mathbb{R}$

c) $f(x) = \frac{7}{2x}, \quad x \in \mathbb{R},\ x \neq 0$

d) $f(x) = \frac{2x+1}{x-3}, \quad x \in \mathbb{R},\ x \neq 3$

e) $f(x) = \frac{4x-9}{2x+1}, \quad x \in \mathbb{R},\ x \neq -\frac{1}{2}$

**2.12 — Question 3.** Given

$$f : x \mapsto \frac{1}{2}x - 5$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) Find $f(12)$ and $f^{-1}(1)$, comment on your result.

**2.12 — Question 4.** For any given one-to-one function, $f$, state the transformation which maps the graph of $y = f(x)$ onto the graph of its inverse function $y = f^{-1}(x)$.

**2.12 — Question 5.** It is given that the graph of the one-to-one function $y = f(x)$ passes through the given point. State the coordinates of the point corresponding to the same point on the curve $y = f^{-1}(x)$.

a) $(1, 3)$

b) $(-5, 4)$

c) $(3, -3e^2)$

d) $(a, b)$

**2.12 — Question 6.** For each of the following functions where $x \geq 0$, solve the equation $f(x) = f^{-1}(x)$.

a) $f(x) = 5x - 2$

b) $f(x) = -x + 4$

c) $f(x) = x^2 - 6$

d) $f(x) = -2x^2 - x + 4$

e) $f(x) = x^3$

f) $f(x) = \frac{2}{x+1}$

**2.12 — Question 7.** Given $f(x) = x^2 + 2x + 5$, explain why $f$ has no inverse.

**2.12 — Question 8.** Given

$$f : x \mapsto -x + 12, \quad x \neq 0$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**2.12 — Question 9.** Given

$$f(x) = \frac{1}{x}, \quad x \neq 0$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**2.12 — Question 10.** Given

$$f(x) = \frac{7x - 48}{x - 7}, \quad x \neq 7$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**2.12 — Question 11.** Given

$$f(x) = \frac{ax - 1}{x - a}, \quad a > 1$$

a) show that $ff(x) = x$

b) Hence write down $f^{-1}(x)$

**2.12 — Question 12. The answers for the following questions which require a range of values should be given in set notation.** Given

$$f : x \mapsto 2x^2 + 1$$

for $x \geq 0$.

a) find $y = f^{-1}(x)$

b) show that there are no solutions to $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**2.12 — Question 13.** Given

$$f(x) = 2e^{3x} - 1$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) given $f(x) = f^{-1}(x)$ has two solutions, sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**2.12 — Question 14.** Given

$$f : x \mapsto 3 + 5^{-x}$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**2.12 — Question 15.** Given

$$f(x) = \frac{2x}{x+5}$$

for $-4 \leq x \leq 1$.

a) find $y = f^{-1}(x)$

b) solve $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**2.12 — Question 16.** For each of the following functions, find the domain and range of $f^{-1}$:

a) $f(x) = 5x - 2$ for $-1 \leq x \leq 5$

b) $f(x) = \ln x$ for $x > 0$

c) $f(x) = e^{x-2}$ for $x \in \mathbb{R}$

d) $f(x) = \sin x$ for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

e) $f(x) = \cos x$ for $0 \leq x \leq \pi$

f) $f(x) = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

**2.12 — Question 17.** A function is given by

$$f(x) = x^4 - x^3 + 2x$$

where $x \in \mathbb{R}$.

a) By using the factor theorem, show that $(x + 1)$ is a factor of $f(x)$

b) Show that $f(x) = 0$ only has two solutions.

c) Explain why an inverse function does not exist.

The domain of $f$ is restricted such that

$$0 \leq x \leq 2.$$

d) Find the domain and range of $f^{-1}$.

**2.12 — Question 18.** Given

$$f(x) = x^2 - 6x + 5$$

for $x \geq p$ where $p$ is a constant.

a) sketch $y = f(x)$ for $x \in \mathbb{R}$

b) state the minimum value that $p$ can be for $f$ to be one-to-one.

For your value of $p$

c) find the range of $f$

d) state the domain and range of $f^{-1}$

e) show that $f(x) = f^{-1}(x)$ only has one solution.

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 19.** Given

$$f(x) = \ln(x - 3)$$

for $x > p$ where $p$ is a constant.

a) State the value of $p$ such that $f$ has the largest possible domain.

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 20.** Given

$$f(x) = \sqrt{x-5}$$

for $x \geq p$ where $p$ is a constant.

a) State the value of $p$ such that $f$ has the largest possible domain.

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

It is known that $x^4 + 10x^2 - x + 30 = 0$ has no solutions.

e) Hence, by considering $f(x) = f^{-1}(x)$, show that there is no intersection between the graphs of $y = f(x)$ and $y = f^{-1}(x)$.

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 21.** Given

$$f(x) = 3 - \frac{1}{x+2}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

For $x > p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) state the coordinates of the point of intersection of the graphs of $y = f(x)$ and $y = f^{-1}(x)$

f) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 22.** Given

$$f(x) = \frac{4x-1}{x-5}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

b) find the range of $f$

c) state the domain and range of $f^{-1}$

**2.12 — Question 23.** For each of the following functions find $f^{-1}(1)$:

a) $f : x \mapsto 5x + 3$

b) $f : x \mapsto x^2 - 8, \quad x \geq 0$

c) $f : x \mapsto \frac{4x+9}{2x+1}, \quad x \neq -\frac{1}{2}$

**2.12 — Question 24.** For

$$f(x) = ax + b$$

where $a$ and $b$ are positive constants, find

a) the gradient function of $f$

b) the gradient function of $f^{-1}$

c) the values of $a$ such that the gradient of $f(x)$ is equal to the gradient of $f^{-1}(x)$

**2.12 — Question 25.** For

$$f(x) = x^2, \quad x \geq 0$$

a) find the gradient of $f(x)$ at $x = 2$

b) find the gradient of $f^{-1}(x)$ at $x = 2$

c) find the value of $x$ such that the gradient of $f(x)$ is equal to the gradient of $f^{-1}(x)$

**2.12 — Question 26.** Given

$$f(x) = -x + 5 \qquad \text{and} \qquad g(x) = 2x - 1$$

for $x \in \mathbb{R}$

a) find $fg(x)$

b) find the value of constant $k$ such that $fg(5) = g^{-1}(k)$

**2.12 — Question 27.** Given

$$f(x) = 2x^2 - 8x \qquad \text{and} \qquad g(x) = 5x + 1$$

for $x \in \mathbb{R}$

a) find $fg(x)$

b) find the value of constant $k$ such that $fg(2) = g^{-1}(k)$

**2.12 — Question 28.** Given

$$f(x) = -2x + 1 \qquad \text{and} \qquad g(x) = 3x + 5$$

$x < 0$ for $f$, $\quad x > 1$ for $g$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**2.12 — Question 29.** Given

$$f(x) = x + 4 \qquad \text{and} \qquad g(x) = x^2 - x$$

$x < 0$ for $f$, $\quad x > 1$ for $g$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**2.12 — Question 30.** Given

$$f(x) = 3x + 1 \qquad \text{and} \qquad g(x) = \frac{2x+2}{x+2}$$

$x \neq -2$ for $g$

a) find $fg(x)$

b) find the domain of $fg$

c) state the range of $fg$

d) find $(fg)^{-1}(x)$

e) state the domain and range of $(fg)^{-1}(x)$

**2.12 — Question 31. The answers for the following questions which require a range of values should be given in interval notation.** Given

$$f(x) = x^2 - 6x + 1$$

for $x \leq 1$.

a) state the range of $f$

b) find $y = f^{-1}(x)$

c) find the solution to $f(x) = f^{-1}(x)$

d) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

e) state the domain and range of $f^{-1}$

**2.12 — Question 32.** Given

$$f : x \mapsto e^{2x} + 2$$

for $x \in \mathbb{R}$.

a) find $y = f^{-1}(x)$

b) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

c) state the domain and range of $f^{-1}$

**2.12 — Question 33.** Given

$$f(x) = \frac{3x}{x-4}$$

for $-5 < x < 2$.

a) find $y = f^{-1}(x)$

b) solve $f(x) = f^{-1}(x)$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes

d) state the domain and range of $f^{-1}$

**2.12 — Question 34.** For each of the following functions, find the domain and range of $f^{-1}$:

a) $f(x) = -2x + 3$ for $-2 \leq x \leq 7$

b) $f(x) = 2\ln x$ for $x > 0$

c) $f(x) = 2 + e^{x+5}$ for $x \in \mathbb{R}$

**2.12 — Question 35.** Given

$$f(x) = \ln(x + 2)$$

for $x > p$ where $p$ is a constant.

a) state the value of $p$

For your value of $p$

b) find the range of $f$

c) find $y = f^{-1}(x)$

d) state the domain and range of $f^{-1}$

e) Given $f(x) = f^{-1}(x)$ has two solutions, sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 36.** Given

$$f(x) = \frac{5x + 2}{x - 9}$$

for $x \neq p$ where $p$ is a constant.

a) state the value of $p$

b) find the range of $f$

c) state the domain and range of $f^{-1}$

**2.12 — Question 37.** Given

$$f : x \mapsto 2x^2 + 4x - 5$$

for $x \geq p$ where $p$ is a constant.

a) state the minimum value that $p$ can be for $f$ to be one-to-one.

For your value of $p$

b) find the domain and range of $f^{-1}$

c) sketch $y = f(x)$ and $y = f^{-1}(x)$ on the same axes.

**2.12 — Question 38.** Given

$$f(x) = 1 + 2x, \quad x < 1 \qquad \text{and} \qquad g(x) = x^2, \quad x > 2$$

a) find $gf(x)$

b) find the domain of $gf$

c) state the range of $gf$

d) find $(gf)^{-1}(x)$

e) state the domain and range of $(gf)^{-1}(x)$

**2.12 — Question 39.** Given

$$f(x) = -2x + 3 \qquad \text{and} \qquad g(x) = \frac{8x + 21}{x + 7}$$

a) find $fg(x)$

b) find the domain of $fg$

c) state the range of $fg$

Given $g$ is now restricted such that $x > -7$

d) find $(fg)^{-1}(x)$

e) state the domain and range of $(fg)^{-1}(x)$

**2.13 — Question 1.** For the functions $f(x) = |x - 2|$ and $g(x) = |x| - 2$, find

a) $f(1)$
b) $f(-3)$
c) $f(0)$
d) $g(1)$
e) $g(0)$
f) $g(-4)$

**2.13 — Question 2.** If $|x| = 4$ find the possible values of $|2x - 5|$.

**2.13 — Question 3.** If $|x| = 2$ find the possible values of $|4x + 1|$.

**2.13 — Question 4.** Write down the $y$-intercept of each of the graphs of the following functions:

a) $y = |x + 2|$
b) $y = |x - 1|$
c) $y = |x - 3| + 5$
d) $y = 2|x + 4| + 8$
e) $y = -5|x - 2| + 1$
f) $y = 6 - 4|x + \frac{1}{2}|$

**2.13 — Question 5.** Write down the vertex of the graph of each of the following functions:

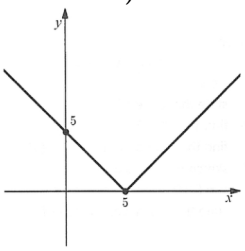
a) $y = |x + 3|$
b) $y = |x - 2|$
c) $y = |x - 1| + 5$
d) $y = |x + 3| - 8$
e) $y = 2|x + 4| + 13$
f) $y = -8|x - 1| + 9$
g) $y = 2 - 7|x + \frac{1}{2}|$

**2.13 — Question 6.** Sketch each of the following, labelling only the vertex and the $y$-intercept:

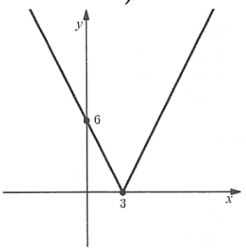
a) $y = |x + 3|$
b) $y = |x - 2| + 4$
c) $y = |x + 1| - 1$
d) $y = |x - 7| - 7$
e) $y = 2|x + 1| + 3$
f) $y = 3|x - 5| + 1$
g) $y = 2|x + 9|$
h) $y = -4|x - 1|$
i) $y = 7 - |x - 6|$
j) $y = 9 - 2|x + 4|$

**2.13 — Question 7.** For each of the following graphs, state the equations:

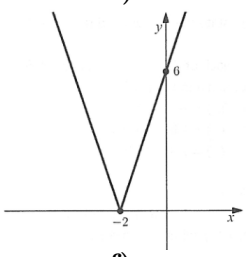
**a)**



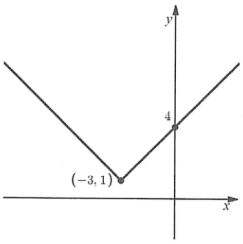

**b)**



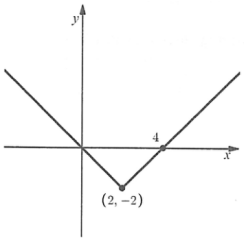

**c)**



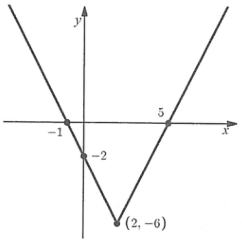

**d)**




**e)**

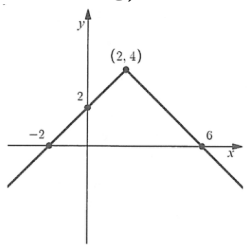



**f)**

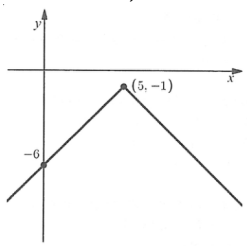



**2.13 — Question 7 (continued).** For each of the following graphs, state the equations:

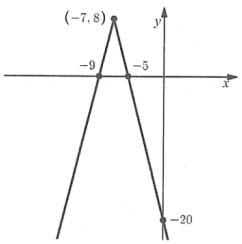
**g)**




**h)**



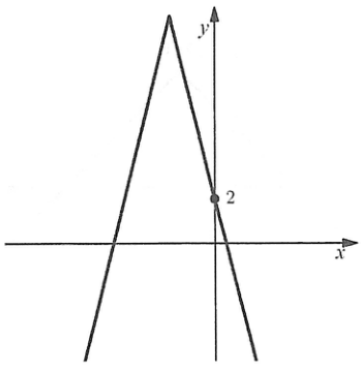

**i)**



**2.13 — Question 8.** Solve each of the following equations:

a) $|x+3|=5$
b) $|x-2|=7$
c) $|2x+1|=4$
d) $-|x+5|=-3$
e) $|x-5|=-3$

**2.13 — Question 9.** The diagram shows a sketch of $f(x)=k-4|x+2|$




Find

a) the value of $k$
b) the values where $f(x)$ intersects with the $x$-axis.

**2.13 — Question 10.** Solve each of the following equations:

a) $|x-1| = |x+8|$
b) $|x+3| = |x-4|$
c) $|x-10| = |x-1|$

**2.13 — Question 11.** Solve each of the following equations:

a) $|x+5| = 2|x+3|$
b) $|x+6| = 3|x-3|$
c) $|x+2| = 5|x+8|$
d) $|x+2| = 10 - |x-5|$
e) $|x-4| - 8 = -|x-3|$
f) $|x+4| = 4 - |x-3|$

**2.13 — Question 12.** Write the following as a modulus inequality:

a) $-6 < x < 6$

b) $-4 < x < 6$

c) $-3 < x < 25$

d) $x < 1, x > 5$

**2.13 — Question 13.** Write $6 \leq x \leq 14$ in the form $|2x - a| \leq b$ where $a$ and $b$ are integers to be found.

**2.13 — Question 14.** Sketch the graph of $y = 6 - |2x - 7|$ and $y = 3$ on the same axes, and hence solve the inequality $6 - |2x - 7| > 3$.

**2.13 — Question 15.** Sketch $y = |2x|$ and $y = |3x - 9|$ on the same axes and hence solve $|2x| \geq |3x - 9|$.

**2.13 — Question 16.** Find the sets of values of $x$ for which, giving your answer in interval notation:

a) $|x+3| > 10$

b) $4 - |x| \leq |x-2|$

c) $|4x| - 2 \geq |7-2x|$

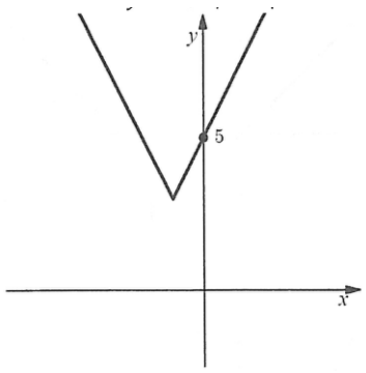
**2.13 — Question 17.** Sketch the graph of $y = |3x + k|$ where $k$ is a positive constant. Include all points of intersection with the coordinate axes.

**2.13 — Question 18.** A function is given to be $f(x) = \frac{7}{8}|x+1| + 4$

a) Sketch $f(x)$

b) Solve $f(x) = 2x + 3$

**2.13 — Question 19.** The diagram shows a sketch of $y = k + 2|x+1|$




a) Find the value of $k$

b) For value of $k$ solve $k + 2|x+1| = x + 8$

**2.13 — Question 20.** Find the values of $x$ such that

a) $|x| + 4 \geq |x + 4|$

b) $-2|x| + 8 > -2|x + 4|$

**2.13 — Question 21.** For the following, find the possible values of $k$ for which there are two distinct solutions:

a) $2|x - 1| + 6 = k$

b) $3|4 - x| - 4 = k$

c) $8 - 2|x + 1| = k$

**2.13 — Question 22.** It is given that $f(x) = |x + 4|$

a) Sketch $f(x)$

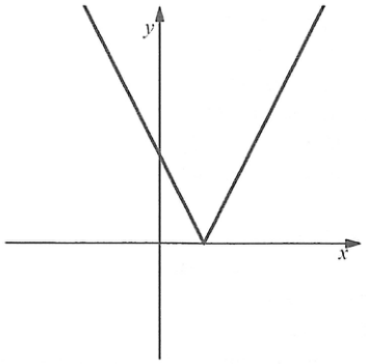
b) Find the values of $k$, such that $f(x) = kx$ has 2 distinct solutions.

**2.13 — Question 23.** It is given that $f(x) = 4 + |x - 2|$

a) Sketch $f(x)$

b) Find the values of $k$ such that $f(x) = 4 + kx$ has 2 distinct solutions.

**2.13 — Question 24.** The diagram shows a sketch of $y = |2x - 5|$




a) State the coordinates of the points of intersection with the axes.
b) The graph of $y = kx + 2$ has two distinct points of intersection, find the possible values of $k$
c) State the solutions of $|2x - 5| = kx + 2$ giving your answer in terms of $k$

**2.13 — Question 25.** Describe the transformations that are required to transform the graph of $y = |x + 3|$ onto the graph of

a) $y = 2|x + 3|$
b) $y = |x - 2|$
c) $y = |x + 5|$
d) $y = |2x + 6|$
e) $y = |2x + 3|$

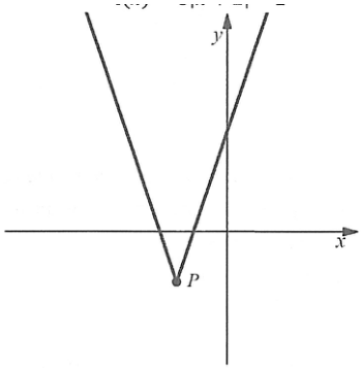
**2.13 — Question 26.** Describe the transformations that are required to transform the graph of $y = |2x - 1|$ onto the graph of

a) $y = 3|2x - 1|$
b) $y = |6x - 3|$
c) $y = |2x - 3|$
d) $y = |2x + 5| + 4$
e) $y = |2x - 4| - 1$

**2.13 — Question 27.** Describe the transformations that are required to transform the graph of $y = |ax + b|$ onto the graph of

a) $y = \frac{1}{2}|ax + b|$
b) $y = |\frac{1}{3}ax + b|$
c) $y = |ax + b + 4a| - 5$

**2.13 — Question 28.** The diagram shows a sketch of $f(x) = 3|x + 2| - 2$




The point $P$ is at the vertex of $f(x)$

a) State the coordinates of $P$

b) Solve the equation $3|x + 2| - 2 = -3x + 18$

A line $y = kx$ where $k$ is a constant.

c) Find the possible values of $k$ such that it intersects $y = 3|x + 2| - 2$ at least once.

**2.13 — Question 29.** A function is given to be $f(x) = k - 3|x + 2|$ where $k \geq 6$

a) Sketch $f(x)$ including labels for all the points of intersection

b) Find the solutions to $k - 3|x + 2| = x + 4$ giving your answer in terms of $k$

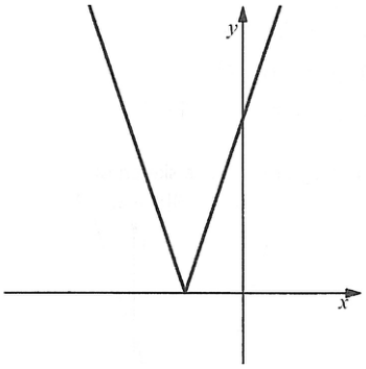
**2.13 — Question 30.**

a) Sketch on a single diagram $y = |x + 2|$ and $y = \frac{k}{x}$ where $k$ is a positive constant.

b) Explain why $|x + 2| = \frac{k}{x}$ has exactly one solution for any positive value of $k$

c) Find the solution to $x|x + 2| = k$ where $k > 0$, giving your answer in terms of $k$

**2.13 — Question 31.** The diagram shows a sketch of $f(x) = 3|x + 2k|$ where $k \geq 0$




Sketch the graph of each of the following, including the coordinate of the vertex and the $y$-intercept in each case

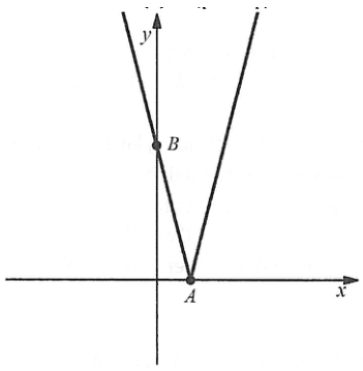
a) $y = -3|x + 2k|$

b) $y = 3|x + 2k| + 4$

**2.13 — Question 32.** Given that $f(x) = k + |x - 2|$ where $k \geq 0$, sketch each of the following, labelling any intersections with the axes:

a) $y = f(x)$
b) $y = -f(x)$
c) $y = 2f(x)$

**2.13 — Question 33.** The diagram shows a sketch of $f(x) = |px + q|$




The point $A$ is at the minimum of the graph and lies on the $x$-axis. The point $B$ is where the graph intersects with the $y$-axis. The line $y = 4x + k$ is parallel to part of the graph and $|px + q| = 4x + k$ has no solutions for $k < -9$. Find the values of $p$ and $q$.

**2.13 — Question 34.** State the number of solutions to the equation $|x+2|+|x|=1$.

**2.13 — Question 35.** State the range each of the following, giving your answer in set notation:

a) $f(x)=|x+1|$
b) $f(x)=-|x+3|$
c) $f(x)=|x-1|+3$
d) $f(x)=2|x+4|-1$
e) $f(x)=5-|x-3|$
f) $f(x)=1-2|x+7|$

**2.13 — Question 36.** It is given that $f(x) = 10|x - 2| - 12$ where $x \geq 0$

a) Sketch $f(x)$

b) State the range of $f(x)$ and domain of $f(x)$

c) Find the values of $x$ such that $f(x) \leq -\frac{1}{2}x + \frac{13}{4}$

**2.13 — Question 37.** It is given that $f(x) = 3|5 - x| + 5$, $x \geq 0$

a) Sketch $f(x)$

b) State the range of $f(x)$

c) Solve $f(x) = \frac{1}{3}x + 15$

**2.13 — Question 38.** For $f(x) = 2x + 1$, $g(x) = |x|$ and $h(x) = -|x|$, find:

a) $fg(x)$
b) $gf(x)$
c) $hf(x)$
d) $hfg(x)$

**2.13 — Question 39.** For $f(x) = x^2 + 1$, $g(x) = |x|$ and $h(x) = -|x|$, find:

a) $fg(x)$
b) $gf(x)$
c) $hf(x)$
d) $fh(x)$
e) $hg(x)$

**2.13 — Question 40.**

a) Sketch on a single diagram $y = |x + 1|$ and $y = k \ln x$ where $k$ is a negative constant.

b) Explain why $|x + 1| = k \ln x$ has exactly one solution for any negative value of $k$

c) Starting with $x_0 = 1$, use the Newton-Raphson method to find the solution to $|x + 1| = -5 \ln x$ giving your answer to 3 significant figures.

**2.13 — Question 41.**

a) Sketch on a single diagram $y = 2|x + 1|$ and $y = ke^x$ where $k$ is a positive constant and $k > 2$

b) Explain why $2|x + 1| = ke^x$ has exactly one solution

c) Starting with $x_0 = -2$, use the Newton-Raphson method to find the solution to $|x + 1| = 2e^x$ giving your answer to 3 significant figures.

**2.13 — Question 42.** Sketch each of the following:

a) $y = 2|x| - 5$
b) $y = 4|x|^2 - 5$
c) $y = 3|x|^2 - 4|x|$
d) $y = 2|x|^2 - 3|x| - 5$
e) $y = -|x|^2 + 4|x| + 5$
f) $y = \ln|x|$
g) $y = \frac{1}{|x|^2}$

**2.13 — Question 43.** Sketch each of the following:

a) $y = |(x-2)(x+5)|$
b) $y = |\ln x|$
c) $y = |\sin x|$ for $-\pi \leq x \leq 2\pi$
d) $y = |\cos x|$ for $-\pi \leq x \leq 2\pi$
e) $y = |x-3| + |x+4|$
f) $y = |x-2| + |2x+1|$
g) $y = |x+5| - |2x+7|$

**2.13 — Question 44.** Sketch the following for $-360° \leq x \leq 360°$:

a) $y = \sin|x|$

b) $y = \tan|x|$

**2.13 — Question 45.** It is given that $f(x) = x^3 + 6x^2 + 3x - 10$. Sketch on separate axes:

a) $y = f(x)$

b) $y = |f(x)|$

c) $y = f(|x|)$

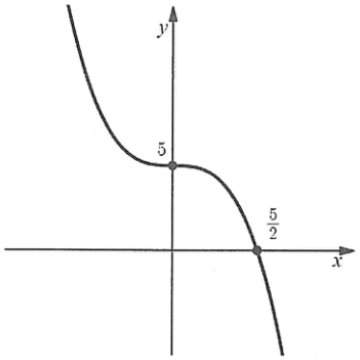
**2.13 — Question 46.** It is given that $f(x) = \cos 2x$ for $-\pi \leq x \leq \pi$. Sketch on separate axes:

a) $y = f(x)$
b) $y = |f(x)|$
c) $y = f(|x|)$

**2.13 — Question 47.** It is given that $f(x) = \tan 3x$ for $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$. Sketch on separate axes:

a) $y = f(x)$
b) $y = |f(x)|$
c) $y = f(|x|)$

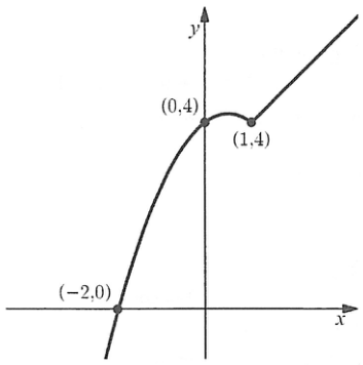
**2.13 — Question 48.** The diagram below shows part of the curve with equation $y = f(x)$




The curve passes through the points $(0, 5)$ and $(\frac{5}{2}, 0)$. Sketch on separate axes:

a) $y = |f(x)|$
b) $y = f(|x|)$
c) $y = 2f(x)$

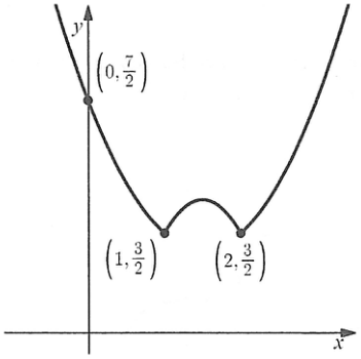
**2.13 — Question 49.** The diagram below shows part of the curve with equation $y = f(x)$




The curve passes through the points $(-2, 0)$, $(0, 4)$ and $(1, 4)$. Sketch on separate axes:

a) $y = |f(x)|$
b) $y = f(|x|)$
c) $y = f(2x)$

**2.13 — Question 50.** The diagram below shows part of the curve with equation $y = f(x)$




The curve passes through the points $(0, \frac{7}{2})$, $(1, \frac{3}{2})$ and $(2, \frac{3}{2})$. Sketch on separate axes:

a) $y = f(|x|)$
b) $y = f(\frac{1}{3}x)$
c) $y = 2f(2x)$

**2.13 — Question 51.** It is given that $f(x) = x^2 + x - 6$

a) Sketch $y = |f(x)|$
b) Sketch $y = f(|x|)$
c) Find the set of values of $x$ for which $f(|x|) = |f(x)|$

**2.13 — Question 52.** It is given that $f(x) = \frac{a}{x}$ for $a > 0$

a) Sketch $y = |f(x)|$

b) Sketch $y = -f(|x|)$

c) Find the set of values of $x$ for which $f(|x|) = |f(x)|$

**2.13 — Question 53.** It is given that $f(x) = e^x - a$ for $a > 0$

a) Sketch $y = f(x)$

b) Sketch $y = |f(x)|$

c) Sketch $y = f(|x|)$

d) Find the set of values of $x$ for which $f(|x|) = |f(x)|$

**2.13 — Question 54.** Find the set of values for which

a) $x^3 = |x|^3$

b) $x^3 - 4x = |x^3 - 4x|$

c) $\ln x = |\ln x|$

d) $e^{x-1} - 4 = |e^{x-1} - 4|$

**2.13 — Question 55.**

a) Write $-(x+2)^2 - 1$ in the form $ax^2 + bx + c$ where $a$, $b$ and $c$ are integers to be found.

b) Sketch $y = -(x+2)^2 - 1$ and $y = -|x|^2 - 4|x| - 5$ on the same axes

c) Hence find the set of value(s) of $x$ such that $-(x+2)^2 - 1 = -|x|^2 - 4|x| - 5$

**2.13 — Question 56.** Solve $x^2 - 4x + 5 = -|x|^2 - 4|x| + 6$.